In [1]:
```
#Exp :- 8
```

In [2]:
```
#Aim:-To perform Operation of Data Visualization
```

In [1]:
```
#Name:Arya patil
#Roll:42
#sec:C
#subject:DSS
#Date:03-09-2026
```

In [2]:
```
import numpy as np
from matplotlib import pyplot as plt
```

In [3]:
```
x=np.arange(1,11)
```

In [4]:
```
x
```

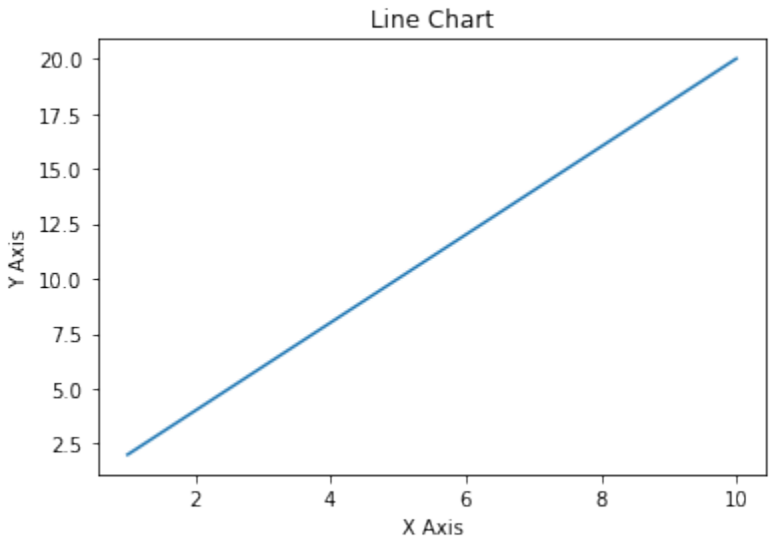
Out[4]:
```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [5]:
```
y=2*x
```

In [6]:
```
y
```

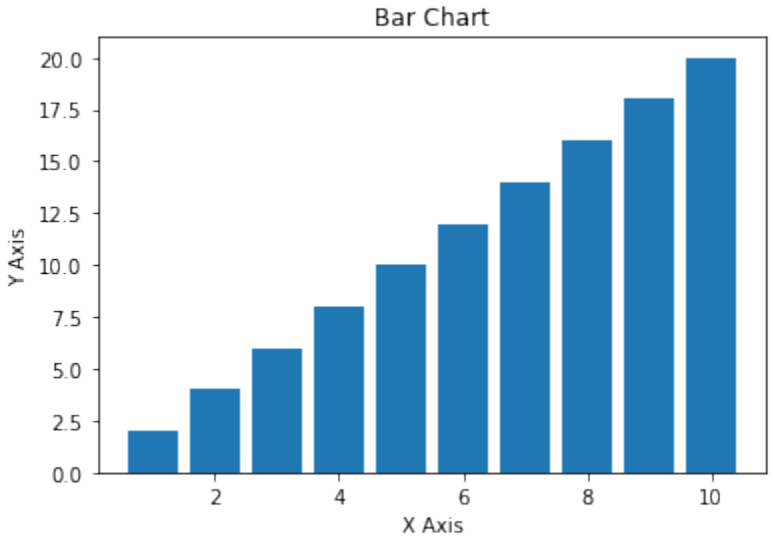
Out[6]:
```
array([ 2,  4,  6,  8, 10, 12, 14, 16, 18, 20])
```

In [7]:
```
plt.plot(x,y)
plt.title("Line Chart")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

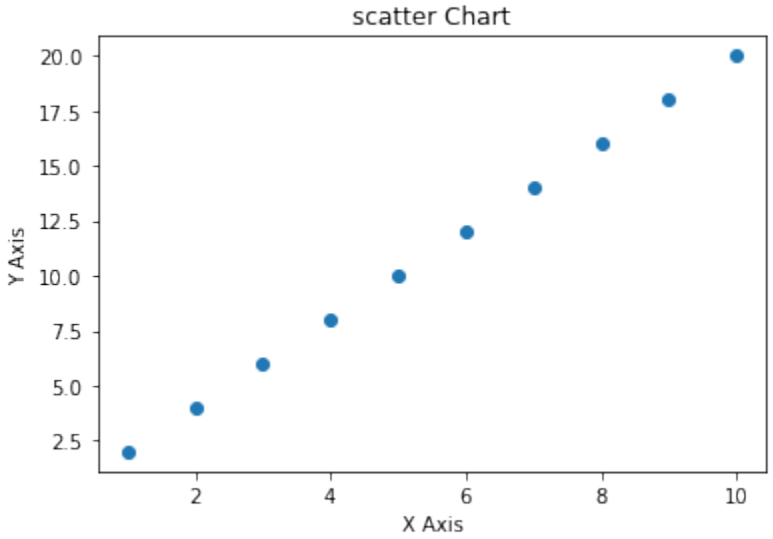


In [8]:
```
plt.bar(x,y)
```

```
plt.title("Bar Chart")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```



In [9]:
```
plt.scatter(x,y)
plt.title("scatter Chart")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

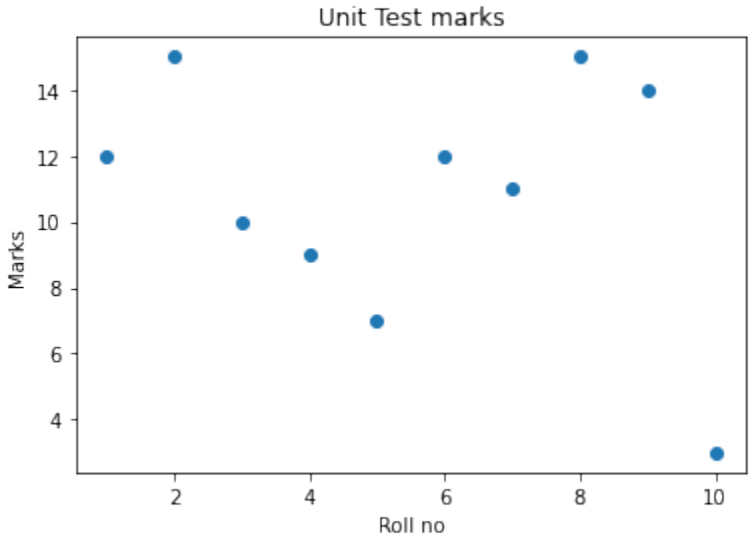


In [10…
```
1 - 12
2-15
3-10
4-9
5-7
6-12
7-11
```

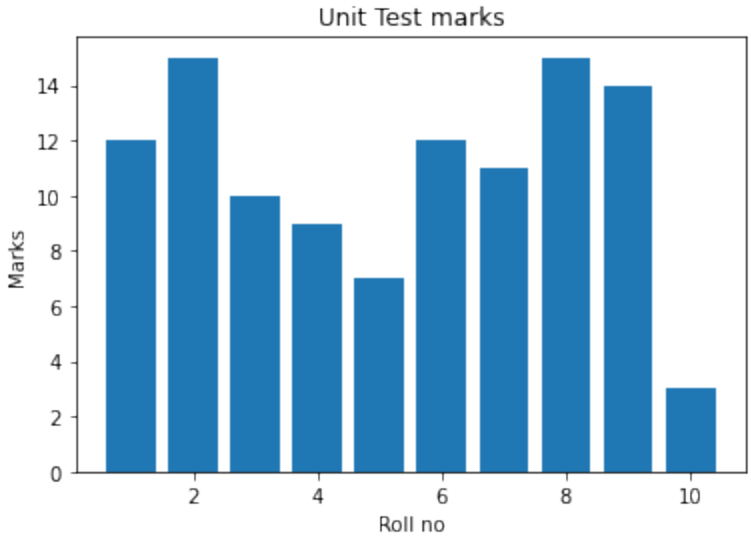
```
8-15
9-14
10-3
```

Out[10]: 7

In [11…
```
R=(1,2,3,4,5,6,7,8,9,10)
M=(12,15,10,9,7,12,11,15,14,3)
plt.scatter(R,M)
plt.title("Unit Test marks")
plt.xlabel("Roll no")
plt.ylabel("Marks")
plt.show()
```



In [12…
```
R=(1,2,3,4,5,6,7,8,9,10)
M=(12,15,10,9,7,12,11,15,14,3)
plt.bar(R,M)
plt.title("Unit Test marks")
plt.xlabel("Roll no")
plt.ylabel("Marks")
plt.show()
```



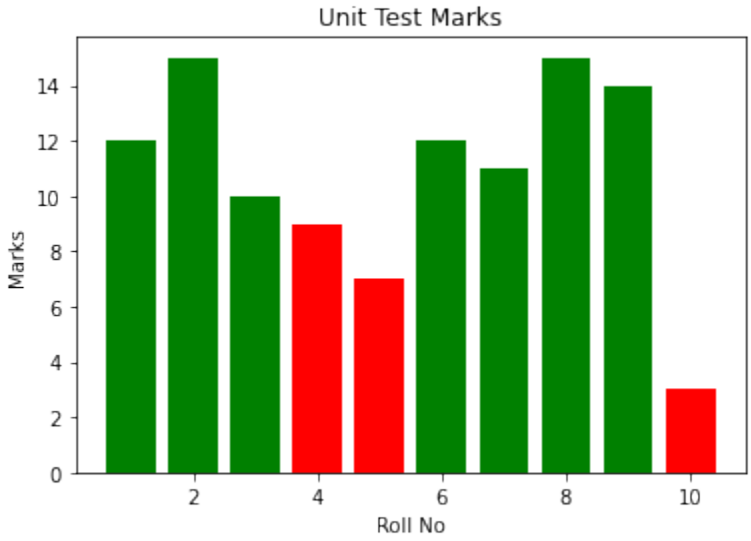
In [13…

```
R = (1,2,3,4,5,6,7,8,9,10)
M = (12,15,10,9,7,12,11,15,14,3)

# Below 10 → Red, 10 or above → Green
colors = ['red' if marks < 10 else 'green' for marks in M]

plt.bar(R, M, color=colors)

plt.title("Unit Test Marks")
plt.xlabel("Roll No")
plt.ylabel("Marks")
plt.show()
```



In [ ]:

In [ ]:

In [ ]:

In [ ]: